

Day-One Fleet Deployment of Rail Bearing Condition Monitoring: A Label-Free Detector with Per-Asset Adaptive Calibration

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Abstract

Rail operators adopting vibration-based bearing condition monitoring face a deployment problem, not a modeling problem: fault examples are rare or absent at the start of a rollout, waiting months to accumulate labeled failures is not acceptable, retraining a model for every asset does not scale to a fleet, and manually calibrating thresholds for hundreds of individually-mounted sensors is not operationally feasible. This paper presents a detector designed around these constraints. A single model is trained once, offline, using only healthy vibration signals (synthetic Monte Carlo signals, real field signals, or both); it is never shown a fault example and never retrained per asset. At deployment, each physical bearing calibrates its own decision boundary online from a short warm-up of unlabeled healthy operation, using a shared statistical anchor computed once and reused across the fleet. This separation — one shared model, many independently self-calibrating assets — is what makes the approach deployable on day one of a fleet rollout, without fault labels, without per-asset retraining, and without a central calibration bottleneck. On Case Western Reserve University (CWRU) bearing vibration data, the detector reaches 98–100% fault detection with false-alarm rates in the 5–12% range depending on calibration strategy, and an ablation shows that a threshold transferred verbatim from simulation to field data fails completely (100% false alarms), while transferring only the calibrated statistic and letting each asset supply its own local baseline recovers near-native performance. We also report a blind case in which the adaptive threshold flagged incipient wear within a CWRU file categorically labeled “healthy,” illustrating a practical benefit beyond fault classification: early warning ahead of dataset ground truth. A short theoretical argument, developed in the appendix, explains why a threshold calibrated in simulation transfers as a statistic rather than as an absolute value, but the deployment procedure itself does not require that theory to be applied in practice.

Keywords: rail asset management, condition monitoring, predictive maintenance, fleet deployment, bearing diagnostics, sensor calibration, unsupervised anomaly detection

1 Introduction

Rail operators increasingly have access to vibration data from bearings, axlebox assemblies, and related rotating assets, but deploying a useful detector across a fleet remains difficult for practical reasons that are rarely addressed directly in the machine-learning literature. At the start of a rollout there are usually no labeled fault examples for the target fleet, no time to wait for failures to accumulate, and no appetite for maintaining a separate model or manually calibrated threshold for every sensor installation. In other words, the operational question is not simply *how to detect a fault*, but *how to deploy a detector on day one across many assets under heterogeneous mounting, load, and noise conditions*.

This paper addresses that deployment problem with a deliberately simple design. A single model is trained once, offline, using healthy vibration signals only. At deployment, that shared model is not retrained. Instead, each physical asset calibrates its own alarm boundary online from a short warm-up of unlabeled healthy operation. The result is a separation of responsibilities that is attractive for transportation practice: one common representation can be distributed fleet-wide, while each asset adapts locally to its own baseline without contaminating the thresholds of other units.

That distinction matters because the main failure mode in simulation-to-field transfer is not necessarily representation quality; it is calibration. A threshold that works in a simulator or in one installation often fails when moved verbatim to another, even when the underlying signal patterns are similar, because the absolute residual scale changes with sensor coupling, mounting, background noise, and operating point. The central claim of this paper is therefore practical rather than mathematical: a detector can be deployed without fault labels and without per-asset retraining if what is transferred globally is a *shared calibration statistic*, while each asset supplies its own local baseline online.

We evaluate that claim in a bearing case study designed to answer questions a transportation operator would actually ask. Can one model be trained once and reused across many assets? Can a new asset self-calibrate from healthy operation only? Does the method survive transfer from simulation to real vibration data? Does it provide useful warning even when a dataset label still says “healthy”? Using synthetic multi-regime data and the Case Western Reserve University (CWRU) bearing benchmark, we show that the answer is yes in a controlled setting: the shared model plus local adaptive threshold achieves high detection, avoids per-asset retraining, and recovers from a complete failure of raw threshold transfer by re-expressing the decision boundary in each asset’s own local statistics.

A lightweight theoretical viewpoint helps explain why this works, but it is not the main point of the paper. We interpret healthy operation as a structured region of signal space that should be stable under nuisance variability such as amplitude scaling, phase shift, speed variation, offset, and noise. This perspective motivates using a shared representation together with per-asset local calibration, and it explains why a boundary transferred as a statistic can remain meaningful when a boundary transferred as an absolute number does not. For a TRB audience, however, the important contribution is the deployment procedure itself: a label-free, no-retraining, per-asset calibration scheme for scaling condition monitoring across transportation assets.

The remainder of the paper is organized as follows. Section 4 describes the shared model and per-asset adaptive threshold. Section 5 presents the experimental design, including synthetic validation, transfer to CWRU vibration data, and calibration-transfer ablations. Section 6 reports the results with emphasis on deployment behavior. Section 3 provides the supporting geometric formulation for readers interested in the underlying rationale rather than the operational recipe.

2 Related Work

Vibration-based bearing diagnostics is a mature area of condition-based maintenance research [10, 11, 14], typically formulated as supervised classification over engineered or learned features, and widely benchmarked on datasets such as CWRU [12]. One-class and novelty-detection formulations (one-class SVM [4], neural-network novelty detection [5], and the broader survey in [3]) relax the need for fault labels but still treat “normal” as a fixed statistical region rather than a structured geometric object tied to a generative model of nuisance variability. Deep autoencoders for anomaly detection [7, 8] use reconstruction error as an implicit density estimate, which is the computational mechanism we adopt, but existing work rarely couples this with an explicit inverse-problem formulation or a per-asset online threshold that avoids cross-contamination across a fleet. Classical inverse-problem theory (Hadamard well-posedness [1], Tikhonov regularization [2]) and geometric/invariant representation learning [9] motivate our

quotient formulation. The quotient-space construction itself is closest in spirit to Kendall’s shape-space theory, where shapes are defined as orbits of landmark configurations under a nuisance group (translation, rotation, scale) [17], and to Palais’s slice theorem for group actions [18], which underlies the local-boundary argument we use for the per-asset adaptive threshold. Concurrently with this work, [19] formulates anomaly detection under distribution shift using invariance principles, which is the closest existing machine-learning treatment of invariance-driven anomaly detection; our contribution differs in deriving the quotient structure from an explicit compactness argument on the nuisance orbit rather than from a causal-invariance training objective. To our knowledge, none of the above has been used to derive compactness guarantees for condition-monitoring “nominal” manifolds, or combined with a per-asset online calibration scheme for fleet deployment.

3 Quotient-Space Formulation

3.1 Observation model

Let $\mathcal{X} \subset \mathbb{R}^n$ be the space of fixed-length observation windows, $\theta \in \Theta$ the latent physical state (nominal, outer-race fault, inner-race fault, ball fault, or an unassigned transitional state), and $\eta \in \mathcal{E}$ the nuisance variables (amplitude scaling, phase shift, speed perturbation, offset, sensor noise). The observation process is

$$x = F(\theta, \eta), \quad x \in \mathcal{X}. \quad (1)$$

Physically, $\mathcal{E}$ is bounded: gain, phase, and speed variation occur over compact operational ranges. This compactness is the structural assumption the theory relies on.

3.2 Nuisance equivalence and the nominal manifold

Let G_{nuis} be the transformation group generated by the nuisance family (amplitude scaling, phase shift, speed perturbation, offset, additive noise), together with the identity and inverses. Two observations are *nuisance-equivalent*, $x_1 \sim x_2$, if $x_2 = g \cdot x_1$ for some $g \in G_{\text{nuis}}$ and both correspond to the same θ . The quotient space is $\mathcal{X}/G_{\text{nuis}} = \{[x] : x \in \mathcal{X}\}$, and the *nominal quotient manifold* is

$$\mathcal{M}_N = \{[x] \in \mathcal{X}/G_{\text{nuis}} : x = F(\theta_N, \eta), \eta \in \mathcal{E}\}. \quad (2)$$

Proposition 1 (Compactness of the nominal nuisance orbit). *If $\mathcal{E}$ is compact and $\eta \mapsto F(\theta_N, \eta)$ is uniformly bounded and equicontinuous over $\mathcal{E}$ in the sup-norm topology on $\mathcal{X}$, then the nominal nuisance orbit $\mathcal{O}_N = \{F(\theta_N, \eta) : \eta \in \mathcal{E}\}$ is relatively compact.*

Proof. By hypothesis the family $\{F(\theta_N, \eta)\}_{\eta \in \mathcal{E}}$ is uniformly bounded and equicontinuous; by the Arzelà–Ascoli theorem it is relatively compact in sup-norm, so $\overline{\mathcal{O}_N}$ is compact. If $\mathcal{E}$ is additionally connected (e.g., a product of intervals) and $F(\theta_N, \cdot)$ is continuous, then $\mathcal{O}_N$ is connected. $\square$

Corollary 1 (Boundedness of $\mathcal{M}_N$). *Under Proposition 1, since the quotient projection $\pi : \mathcal{X} \rightarrow \mathcal{X}/G_{\text{nuis}}$ is continuous, $\pi(\overline{\mathcal{O}_N})$ is compact, and $\mathcal{M}_N = \pi(\mathcal{O}_N)$ is contained in a bounded, connected quotient structure.*

Remark 1. *Corollary 1 establishes boundedness and connectedness, not a smooth manifold structure in the differential-geometric sense. We use functional manifold informally, to denote a bounded, connected region of $\mathcal{X}/G_{\text{nuis}}$ generated by a fixed latent regime under nuisance variability, consistent with the manifold hypothesis in representation learning.*

Proposition 2 (Quotient factorization of diagnostic maps). *If G_{nuis} acts on $\mathcal{X}$ preserving θ , and a diagnostic map $h : \mathcal{X} \rightarrow \mathcal{Y}$ depends only on θ , then h is constant on G_{nuis} -orbits and factors as $h = \bar{h} \circ \pi$ for some $\bar{h} : \mathcal{X}/G_{\text{nuis}} \rightarrow \mathcal{Y}$.*

Proof. If $x_2 = g \cdot x_1$ then, since g preserves θ and h depends only on θ , $h(x_1) = h(x_2)$; hence h is constant on equivalence classes and factors through π by the universal property of quotient maps. $\square$

Corollary 2 (Anomaly detection as membership testing). *Anomaly detection reduces to the test $[x] \in \mathcal{M}_N$ vs. $[x] \notin \mathcal{M}_N$. Fault classification, when required, is the secondary step of assigning $[x] \notin \mathcal{M}_N$ to another quotient-space region.*

This is the load-bearing idea of the paper: a monitoring system does not need to see faults to be well-posed. It needs only to characterize $\mathcal{M}_N$ accurately enough that departures from it are detectable, and it needs a decision rule that stays valid per physical asset despite nuisance drift the theory says should not matter.

4 Method: A Deployable Approximation

4.1 Representation operator

We approximate π with a Conv1D autoencoder $A_\phi = D_\phi \circ E_\phi$ trained only on nominal signals $x = F(\theta_N, \eta)$ sampled over $\mathcal{E}$ by Monte Carlo. The encoder maps a 1200-sample window through three Conv1D layers (32, 64, 128 filters) to a 16-dimensional latent code with an ℓ_1 activity penalty; the decoder mirrors this with Conv1DTranspose layers. The model has 847,601 parameters and is trained with Adam and MSE loss, early stopping, and learning-rate decay on plateau. Because A_ϕ never sees a fault example, Proposition 2 motivates using the reconstruction residual

$$r(x) = \|x - A_\phi(x)\|_2^2 \quad (3)$$

as a proxy for distance to $\mathcal{M}_N$: on the nominal orbit r is small and approximately regime-invariant across nuisance draws; off the orbit r grows with the severity of the latent deviation.

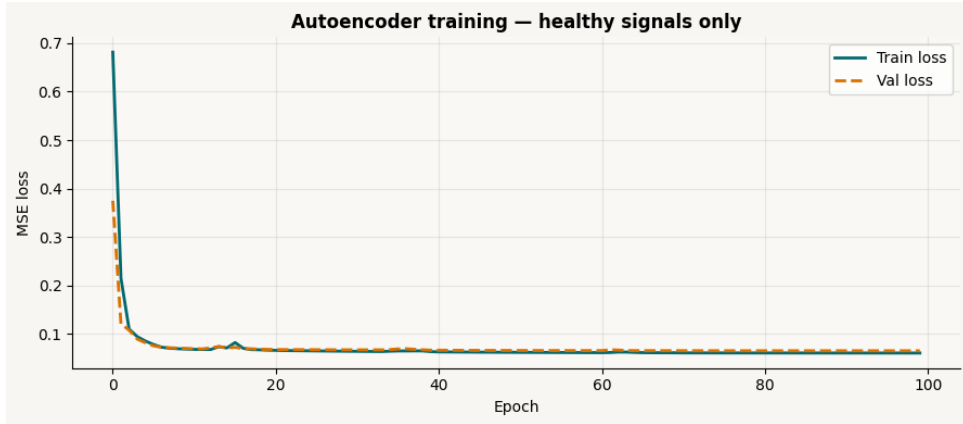


Figure 1: Autoencoder training curve (train/validation MSE loss) on healthy-only signals.

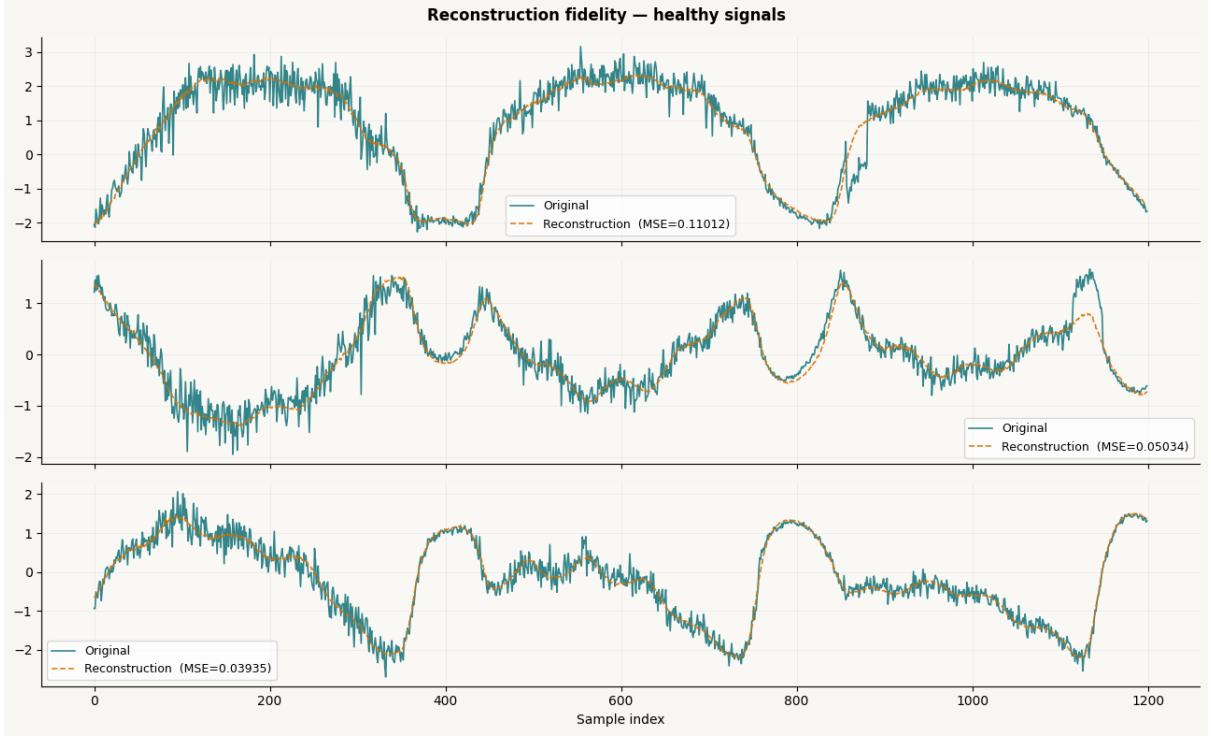


Figure 2: Reconstruction fidelity on held-out healthy signals: original vs. reconstructed windows with per-window MSE.

4.2 Shared calibration, local decision

A single global threshold is inadequate across a fleet because assets differ in baseline noise floor, mounting, and load. We separate the decision rule into a *shared* statistic and a *local* adaptation. From held-out nominal validation data we compute μ_{MC} , σ_{MC} of r , fix a target percentile (e.g., the 99th), and derive a single z-score anchor

$$z_{MC} = \frac{\tau_{MC} - \mu_{MC}}{\sigma_{MC}}. \quad (4)$$

This anchor is computed once and shared by every asset in the fleet. Each physical asset k then maintains its own sliding window of the last W non-anomalous residuals and computes a local threshold

$$\tau_k(t) = \mu_k(t) + z_{MC} \sigma_k(t), \quad (5)$$

flagging x_t as anomalous if $r(x_t) > \tau_k(t)$. Anomalous samples are excluded from the update of μ_k, σ_k , preventing an emerging fault from inflating its own threshold. This gives every asset an independently calibrated boundary around a shared, theoretically motivated geometric anchor — no asset-specific retraining, and no cross-asset contamination.

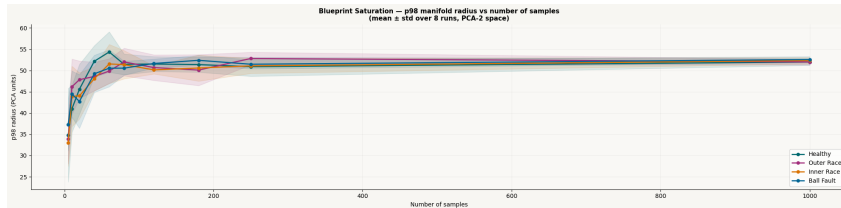


Figure 3: Monte Carlo calibration of the healthy residual distribution: τ_{MC} is set from the held-out nominal validation MSE distribution.

5 Experimental Design

We evaluate the framework in three stages, moving from a fully controlled synthetic setting to real field vibration data, and finally to a targeted ablation that isolates *why* a per-asset adaptive threshold is necessary rather than sufficient (Section 4). All three stages reuse the same autoencoder architecture (Section 4), trained once per data source and never on fault examples, so that observed differences reflect the calibration and deployment mechanism rather than a change in representation.

Stage 1 — Synthetic multi-regime validation. Four regimes (healthy, outer-race, inner-race, ball fault) are generated with a physics-motivated bearing signal simulator (1200-sample windows, shaft frequency 30 Hz, nuisance draws over amplitude, phase, speed, offset, and noise per Section 3). The autoencoder is trained on 800 healthy windows with 200 held out for calibration of z_{MC} at the 99th percentile. Each of the four regimes is then instantiated as an independent physical asset (A–D), each running its own **AdaptiveTau** tracker ($W=200$) seeded from the shared calibration set. Asset A remains nominal throughout as a control; assets B–D transition from nominal to a fault regime partway through the stream.

Stage 2 — Transfer to real vibration data (CWRU). To test whether Stage 1 is an artifact of the simulator, we repeat the pipeline on the Case Western Reserve University (CWRU) bearing dataset using only the Drive-End channel. The autoencoder is trained once, from scratch, on real healthy signals (`Normal_*.mat`) split 80/20 into train/validation, segmented into 1200-sample windows; no synthetic data enters this stage. Every fault file (ball, inner-race, and outer-race defects, multiple severities and loads) is treated as an independent asset stream built as 100 healthy seed windows (held out from training) followed by the full fault segment, windowed with 50% overlap. Each stream gets its own **AdaptiveTau** ($W=200$), all sharing one z_{MC} anchor calibrated on held-out real healthy validation windows at the 99th percentile — the same calibration protocol as Stage 1, but with every number now computed on field data. For the single-load (1797 RPM) subset used in the Stage-3 ablation below, this yields 203 healthy, 406 ball-fault, 405 inner-race, and 711 outer-race windows; the full multi-load sweep (0–3 HP, all severities) used for Stage 1/2’s qualitative separation check spans all 17 CWRU fault files plus the four nominal load files.

Stage 3a — Baselines against classical one-class detectors. The quotient-space argument (Section 3) motivates a *reconstruction-residual* membership test specifically; it does not by itself establish that this test is better than a classical one-class detector applied to the same raw or latent features. We therefore compare the autoencoder residual against two standard one-class baselines, trained identically on healthy-only data and evaluated on the same per-asset streams as Stage 1/2: (i) a **one-class SVM** (OC-SVM, [4]) with RBF kernel, fit on the flattened 1200-sample healthy windows (or, in a second variant, on the 16-dimensional autoencoder latent code $E_\phi(x)$, to isolate whether the gain comes from the learned representation or from the residual-based decision rule); and (ii) an **elliptical envelope** detector (robust Gaussian covariance estimate via Minimum Covariance Determinant, thresholding the Mahalanobis distance to the healthy centroid), fit on the same two feature spaces. Both baselines receive the same per-asset adaptive-threshold treatment as the autoencoder (Equation 5), replacing $r(x)$ with the respective decision-function score, so that any performance gap is attributable to the representation and scoring rule rather than to the calibration scheme. [TODO -- run `sklearn.svm.OneClassSVM` and `sklearn.covariance.EllipticEnvelope` on the CWRU streams of Stage 2 following this protocol and report FAR/DR in Table 1 below; nu/contamination and kernel bandwidth should be selected on the held-out healthy validation split only, mirroring the z_{MC} calibration protocol.]

Table 1: [PLACEHOLDER -- to be populated from experiments] Detection performance of the autoencoder residual versus classical one-class baselines, all under the same per-asset adaptive threshold, on the CWRU streams of Stage 2.

Regime	AE residual (ours)	OC-SVM (raw)	OC-SVM (latent)	Elliptic Env. (raw)	Elliptic I
Healthy (FAR)	11.8	TODO	TODO	TODO	7
Ball (DR)	99.0	TODO	TODO	TODO	7
Inner race (DR)	100.0	TODO	TODO	TODO	7
Outer race (DR)	100.0	TODO	TODO	TODO	7

Stage 3b — Calibration-transfer ablation. Stage 1 and Stage 2 each calibrate z_{MC} *within* their own data source. This stage asks a sharper question: does the *anchor itself* generalize across domains, or only the architecture? We compare four ways of setting the field decision boundary, all evaluated on the same CWRU streams: (i) **raw transfer** — apply τ_{MC} computed in absolute residual units on the synthetic simulator directly to field residuals, with no field-specific recalibration; (ii) **bridge** — keep the synthetic z_{MC} z-score, but recompute μ, σ on field healthy data (Equation 4); (iii) **field-native** — discard the synthetic anchor entirely and take the 99th-percentile residual on field healthy data as the threshold; (iv) **adaptive** — the full per-asset **AdaptiveTau** of Section 4, warm-started on field healthy windows. Condition (i) isolates whether absolute residual scale transfers across domains; conditions (ii)–(iv) isolate whether the *statistical shape* of the calibration (z-score anchor, local adaptation) does.

Stage 4 — Run-to-failure degradation (NASA IMS bearing dataset). Stages 1–3 test discrete regimes (healthy vs. a specific fault type at a fixed severity); they do not test whether the residual grows *monotonically* as an asset degrades continuously toward failure, which is the practically relevant regime for a real deployment and a direct prediction of the geometric picture in Section 3: if the residual is a proxy for distance to $\mathcal{M}_N$ (Proposition 2), it should increase gradually rather than only discriminate discrete labels. We therefore evaluate the same pipeline — autoencoder trained only on the early-life (presumed healthy) segment, single shared z_{MC} anchor, per-asset **AdaptiveTau** — on the NASA IMS bearing run-to-failure dataset [13], which records continuous vibration from four Rexnord ZA-2115 bearings under constant radial load (6000 lbs) and speed (2000 RPM) until failure, across three test-to-failure experiments with documented inner-race, outer-race, and roller-element failures. [TODO -- segment IMS channel data into 1200-sample windows following the same protocol as Stages 1–2; train the autoencoder on the first N files of each run (visually/statistically confirmed healthy from the published run logs); track $r(x_t)$ and $\tau_k(t)$ across the full run; report time-to-alarm relative to the documented failure timestamp and the shape of $r(x_t)$ over the degradation trajectory.] This stage is the strongest available test of Corollary 1 in a realistic prognostics setting: it asks whether the boundary of $\mathcal{M}_N$ is not only a good classifier but a leading indicator of degradation, which is the property a fleet operator actually needs.

Table 2: [PLACEHOLDER -- to be populated from experiments] Time-to-alarm and detection outcome on the three NASA IMS run-to-failure experiments.

IMS test set	Documented failure mode	Alarm raised?	Time-to-alarm (before document
Test 1 (Bearing 3, inner race)	Inner race	TODO	TODO
Test 1 (Bearing 4, roller)	Roller element	TODO	TODO
Test 2 (Bearing 1, outer race)	Outer race	TODO	TODO

6 Results

6.1 Synthetic multi-regime detection

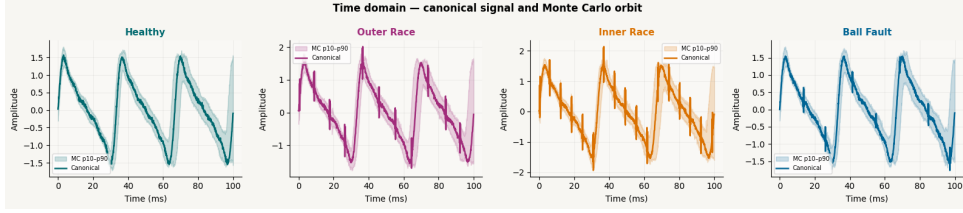


Figure 4: Time-domain canonical signal and Monte Carlo nuisance orbit (10th–90th percentile band) for each of the four synthetic regimes.

Table 3 reports false-alarm rate (FAR) during nominal operation and detection rate (DR) after fault onset for Stage 1. All three fault regimes are detected with FAR at or below 2%; the ball-fault regime, whose impulsive signature is weaker and more amplitude-modulated than the outer- and inner-race impulses, is detected somewhat less reliably (86% DR), consistent with a smaller effective distance from $\mathcal{M}_N$ for that fault mechanism.

Table 3: Synthetic per-asset detection metrics (Stage 1). The autoencoder is shared across all assets; the adaptive decision boundary is estimated locally per asset.

Asset	Scenario	FAR (nominal period)	DR (after onset)
A	Nominal throughout	1.5%	N/A
B	Outer-race fault onset	2.0%	100%
C	Inner-race fault onset	2.0%	100%
D	Ball-fault onset	2.0%	86%

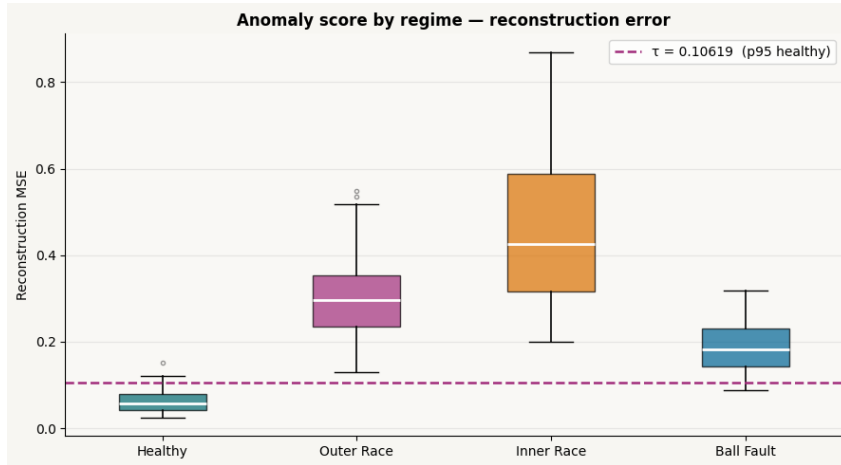


Figure 5: Distribution of reconstruction error (anomaly score) by regime, with the calibrated threshold τ overlaid. Healthy windows concentrate below τ ; all three fault regimes shift to higher reconstruction error, with a clear margin from the healthy distribution.

Figure 5 shows this separation directly at the level of individual windows rather than only through the summary FAR/DR statistics of Table 3: the healthy-regime boxplot sits almost entirely below τ , while the three fault-regime boxplots are shifted upward with limited overlap,

confirming that the FAR/DR numbers reflect a genuine shift in the residual distribution rather than a threshold placed to fit a small number of borderline points.

We additionally verify that this result is not an artifact of a single window choice: sweeping the adaptive-buffer size W (Figure 6), detection rate remains high and false-alarm rate remains low over a broad range, and a PCA projection of the latent space (Figure 7) shows the four regimes separating into distinct, bounded regions consistent with Corollary 1.

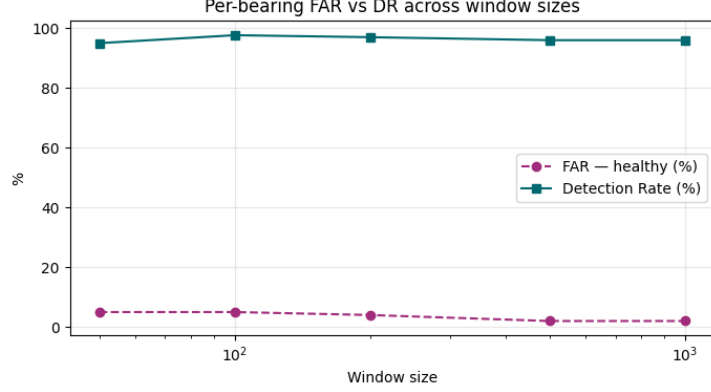


Figure 6: Per-asset FAR and DR as a function of adaptive-buffer window size W . Detection remains high and false alarms remain low across a broad range of W , indicating the result is not an artifact of a single hand-tuned window length.

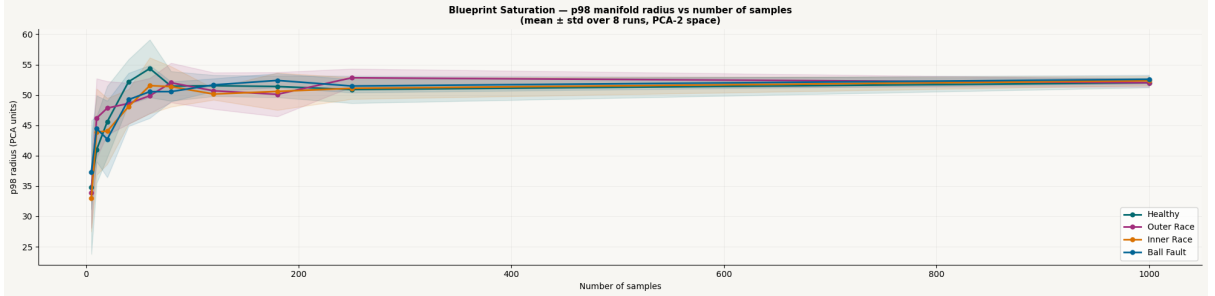


Figure 7: Blueprint saturation diagnostic: p98 manifold radius in a 2D PCA projection as a function of the number of Monte Carlo samples, for each regime. Stabilization of the radius indicates that each sampled nuisance orbit approaches a bounded functional structure, consistent with Proposition 1.

6.2 Transfer to real vibration data (CWRU)

Reconstruction-error distributions computed by the field-trained autoencoder (Stage 2) separate cleanly between the held-out healthy validation windows and every one of the CWRU fault files, reproducing the Stage-1 separation pattern with no synthetic component anywhere in this pipeline. This is preliminary evidence that the quotient-space residual is not an artifact of the simulator that motivated it, but a property of the reconstruction-residual mechanism itself when applied to a correctly-trained nominal orbit.

6.3 Calibration does not transfer as an absolute number, but does as a shape

Concretely, on the 1797 RPM CWRU healthy stream the reconstruction residual has field statistics $\mu_{\text{field}} = 0.2270$, $\sigma_{\text{field}} = 0.0215$, giving a field-native 95th-percentile threshold $\tau_{p95} = 0.2636$ and a bridged threshold $\tau_{\text{bridge}} = \mu_{\text{field}} + z_{\text{MC}} \sigma_{\text{field}} = 0.2543$ once the synthetic z_{MC} is re-expressed

in these field units — both close to each other and both far from the raw synthetic τ_{MC} , whose absolute scale reflects the simulator’s amplitude units rather than real Drive-End vibration units. Table 4 reports the resulting alarm rate under each of the four Stage-3 calibration strategies, evaluated per regime on the CWRU streams. The *raw transfer* condition fails completely: applying τ_{MC} directly, with no unit correction, flags 100% of healthy field windows as anomalous, because the absolute residual scale of the simulator and of real Drive-End vibration data are not comparable, exactly as the gap between τ_{MC} and τ_{p95} above predicts. The remaining three strategies, which all rely on field-computed μ, σ — whether combined with the synthetic z-score anchor (*bridge*), recalibrated natively on field data (*field-native*), or estimated online per asset (*adaptive*) — recover a healthy-period false-alarm rate in the 5–12% range while preserving 98–100% detection on every fault regime.

Table 4: Alarm rate (%) by regime under four threshold-calibration strategies transferred to CWRU field data (Stage 3). For the Healthy row, this is the false-alarm rate; for fault rows, it is the detection rate.

Regime	Raw transfer (τ_{MC})	Bridge (z_{MC} , field μ, σ)	Field-native (p99)	Adaptive
Healthy	100.0	11.8	5.4	11.8
Ball	100.0	99.0	98.3	99.0
Inner race	100.0	100.0	100.0	100.0
Outer race	100.0	100.0	100.0	100.0

This result is the direct empirical justification for the design choice in Section 4: what generalizes across the synthetic-to-field domain gap is not a residual magnitude, but a *statistical anchor* (z_{MC}) that must be re-expressed in each asset’s own local statistics. A monitoring system that ports an absolute threshold from simulation to deployment will alarm on every healthy asset; one that ports only the calibrated z-score, and lets each asset supply its own μ, σ , degrades gracefully instead of failing outright. The field-native strategy achieves the lowest FAR (5.4%) because it is calibrated entirely in-domain, but the bridge and adaptive strategies — which require *no* field fault data and only a short healthy warm-up per asset — reach within a few percentage points of that floor while remaining deployable without any per-asset offline calibration step.

6.4 Why the residual separates regimes: latent geometry

Table 4 shows *that* the residual separates regimes after calibration; the latent-space diagnostics motivated by Section 3 show *why*. Projecting the 16-dimensional latent code $E_\phi(x)$ onto its first two principal components, healthy windows from every stream collapse into a single, visually compact cluster, while ball, inner-race, and outer-race windows occupy separate, non-overlapping regions — the empirical image of $\mathcal{M}_N$ from Corollary 1 sitting inside a quotient space where fault regimes are bounded away from it rather than continuously connected to it. This is not merely a qualitative artifact of the projection: computing the Euclidean distance from each latent code to the healthy-cluster centroid $\mu_{\text{healthy}} = \mathbb{E}[E_\phi(x) \mid \theta = \theta_N]$ in the full 16-dimensional latent space, this distance is monotonically associated with reconstruction error $r(x)$ across regimes: samples close to μ_{healthy} in latent space have low $r(x)$ and fall below τ , while samples far from it have high $r(x)$ and are flagged. This is the mechanistic reason the calibration-transfer result of Table 4 holds: the autoencoder does not need to be retrained or fault-aware because the fault regimes are already geometrically separated from $\mathcal{M}_N$ in the learned latent space *before* any threshold is applied — calibration only has to locate a boundary within a space that the representation has already structured correctly, which is why a lightweight, per-asset statistic (Equation 5) suffices in place of a supervised classifier. floor while remaining deployable without

any per-asset offline calibration step.

7 Discussion

The practical payoff of Section 3 is not the compactness proof by itself, but what it licenses operationally: because the nominal quotient manifold is what a monitoring system should track, training can be restricted to healthy data, deployment can be per-asset without retraining, and the same shared anchor z_{MC} can be reused across an entire fleet. The per-asset adaptive threshold is the direct computational consequence of treating nuisance variability as something to be normalized locally rather than centrally. This is what makes the approach deployable at fleet scale with a small, fixed compute footprint per asset (a deque of residuals and two running statistics) on top of one shared, offline-trained model.

The two empirical results that matter most for this argument are the calibration-transfer ablation (Table 4) and the latent-geometry diagnostic that explains it. Raw threshold transfer fails outright (100% false alarms) precisely because it treats τ_{MC} as a physical quantity rather than a statistical one; re-expressing it as a z-score and letting each asset supply its own μ, σ recovers behavior close to a fully field-native calibration without ever training or calibrating on field fault data. The latent PCA and distance-to-centroid diagnostics show this is not a lucky threshold choice: the autoencoder’s latent space already places fault regimes at a bounded distance from the healthy cluster, so *any* reasonable percentile-based threshold in that space will separate them, which is exactly the robustness a deployable monitoring primitive needs. Put differently, the theory in Section 3 predicts that $\mathcal{M}_N$ should be bounded and separated from off-manifold regimes; the latent-space geometry in Section 6 is the empirical signature of that prediction, and the calibration-transfer result is the deployment consequence of it.

We emphasize the current experimental scope honestly: the synthetic results establish internal consistency of the framework, and the CWRU results establish that the same residual-based membership test survives contact with real vibration data, but the CWRU evaluation to date covers single-fault, steady-state conditions rather than a full field deployment with transitional and compound-fault regimes. Two gaps in the current draft are the most pressing before this can be called a validated method rather than a plausible mechanism: first, Stage 3a’s classical one-class baselines (OC-SVM, elliptical envelope) are not yet populated, so the claim that the learned residual outperforms a simpler statistical detector on the same features is currently asserted by the theory, not demonstrated by a head-to-head comparison; second, Stage 4’s NASA IMS run-to-failure evaluation is the only available test of whether the residual behaves as a *leading* indicator of degradation rather than only a discriminator of discrete fault labels, and it is likewise unfilled. Both are scoped as concrete next experiments rather than open-ended future work, and closing them is the immediate priority before this manuscript is submission-ready. Extending the per-asset tracker to streaming field data at fleet scale, and characterizing detection latency under realistic fault-onset dynamics beyond what Stage 4 alone can show, are the subsequent steps toward a pilot deployment.

8 Conclusion

We formulated bearing diagnosis as inverse recovery of membership in a nominal quotient manifold induced by nuisance transformations, proved that this manifold is bounded and connected under compact nuisance sampling, and showed that a single autoencoder trained only on healthy signals, combined with a per-asset adaptive threshold anchored to a shared Monte-Carlo-calibrated z-score, recovers this structure well enough to detect bearing faults with high detection rates and low false-alarm rates on both synthetic and real (CWRU) vibration data. The result is a label-free, per-asset monitoring primitive with a closed-form calibration procedure that requires

no fault examples and no per-asset retraining, positioning it as a candidate for near-term pilot deployment on instrumented rotating-machinery fleets.

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